

CUADebugger Pipeline

Inspect → Localize → Diagnose → Re-rollout

1 Failed trajectory package

TASK Add unpaid students' e-mails to the receiver field
APPS Calc → Thunderbird

Input: screenshots + reasoning/action trace

t=5 payment record



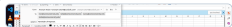
t=6 **ROOT** invents Kelly; misses Fox



t=10 wrong recipients

Iron, Kelly, Nancy, Stella; no Fox

t=12 **FAIL** declares success; reward 0



INPUT task + trajectory + terminal failure

2 Tool-guided inspection & localization

get_step_details(6) +
get_step_details(10)

t=6: before



t=6

Kelly / no
Fox

t=10: result



before / after

response +
action

later evidence

Candidate inspection path

inspect
t=6

compare
t=5/6

inspect
t=10

localize
t=6

Root: t=6 reasoning invents Kelly and omits Fox.

OUTPUT root-step candidate + grounded evidence

3 Structured RCA record

Input: candidate step + inspected evidence

ROOT STEP root_error_step = 6
Invents Kelly; omits Fox

TAXONOMY taxonomy_tag = P1
Visual hallucination

EVIDENCE t=5: Fox = No; t=6: Kelly/no Fox;
t=10: wrong list

CORRECTION Use all No rows: Fox, Iron,
Nancy, Stella

OUTPUT structured RCA record

4 Debugger-guided re-rollout

Matched continual: Claude 4.5 Sonnet;
restart t=51

Continual re-rollout: Recovery / Overall

- ✓ Direct continuation: 12.50% / 61.22%
- ✓ Self-Debug: 13.89% / 61.77%
- ✓ Machine RCA: 21.53% / 64.82%
- ✓ Ours + episodic memory: 25.69% / 66.48%

Takeaway

RCA outperforms reflection; episodic memory adds **+4.16 pp recovery** (**+1.66 pp overall**).

OUTPUT re-rolled trajectory + outcome

Optional Episodic Memory (+4.16 pp recovery / +1.66 pp overall)

1 · Distill failures

Trigger · failed action · correction · evidence

2 · Build query

Task · app · state · suspected action

3 · Retrieve

Top-k transferable lessons

4 · Verify

Check screenshots and trace; discard mismatches